Advanced SOC Concepts: Log Analysis, Threat Intelligence & Incident Escalation

- **Theoretical Knowledge:-**

1. Introduction:-

In this report, I studied important concepts used in a Security Operations Center (SOC). These include log analysis, threat intelligence integration, and incident escalation. These concepts help in detecting cyber attacks, analyzing threats, and responding effectively.

2. Advanced Log Analysis:-

- Log analysis is the process of examining system logs to identify suspicious activities.
- One important concept is *log correlation*. It means connecting logs from different sources like firewalls, servers, and applications. For example, multiple failed login attempts (Event ID 4625) followed by unusual network activity may indicate an attack.
- Another concept is *anomaly detection*. It helps in identifying unusual behavior such as login at odd hours or large data transfer. These activities may indicate a security threat.
- Log enrichment is also important. It adds extra information like IP location or user details to logs. This helps in better understanding of the activity.
- From this, I learned how to analyze logs and detect attacks effectively.

3. Threat Intelligence Integration:-

- Threat intelligence helps in identifying known threats using data from external sources.
- There are different types of intelligence like *IOCs (Indicators of Compromise)* such as malicious IPs, file hashes, and URLs. These help in detecting known attacks.

- Another concept is *TTPs (Tactics, Techniques, Procedures)*, which explain how attackers behave. These are defined in MITRE ATT&CK framework.
- Threat intelligence can be integrated into SIEM tools to automatically match suspicious activity with known threats. For example, checking an IP address in VirusTotal to see if it is malicious.
- This helps in faster detection and better decision-making.

4. Incident Escalation Workflows:-

Incident escalation is the process of handling security incidents based on their severity. SOC teams are divided into levels:

- *Tier 1*: Basic monitoring and triage.
- *Tier 2*: Detailed investigation.
- *Tier 3*: Advanced threat analysis.

If an incident is serious, it is escalated from Tier 1 to Tier 2.

Communication is also important. Reports like *SITREP (Situation Report)* are used to explain the issue clearly.

Automation tools like SOAR can help in automatically assigning incidents and reducing manual work.

This helps in quick and efficient response to cyber threats.

• Practical Application:-

5. Introduction:-

In this practical work, I performed various SOC operations such as log analysis, threat intelligence integration, alert triage, incident response, and evidence collection. The aim was to understand how real-world security monitoring and response works using tools like Kibana, VirusTotal, and Wazuh.

6. Advanced Log Analysis:-

- In this task, I analyzed logs using Kibana to detect suspicious activities.
- First, I performed log correlation by analyzing failed login attempts (Event ID 4625) and linking them with outbound traffic. This helped in identifying possible brute-force attacks.

- Then, I applied anomaly detection by observing unusual patterns such as high data transfer and repeated login failures.
- I also used log enrichment by adding GeoIP data to identify the location of IP addresses.

- **Log Correlation Table:-**

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-03-31 12:00:00	4625	192.168.1.100	8.8.8.8	Failed login detected

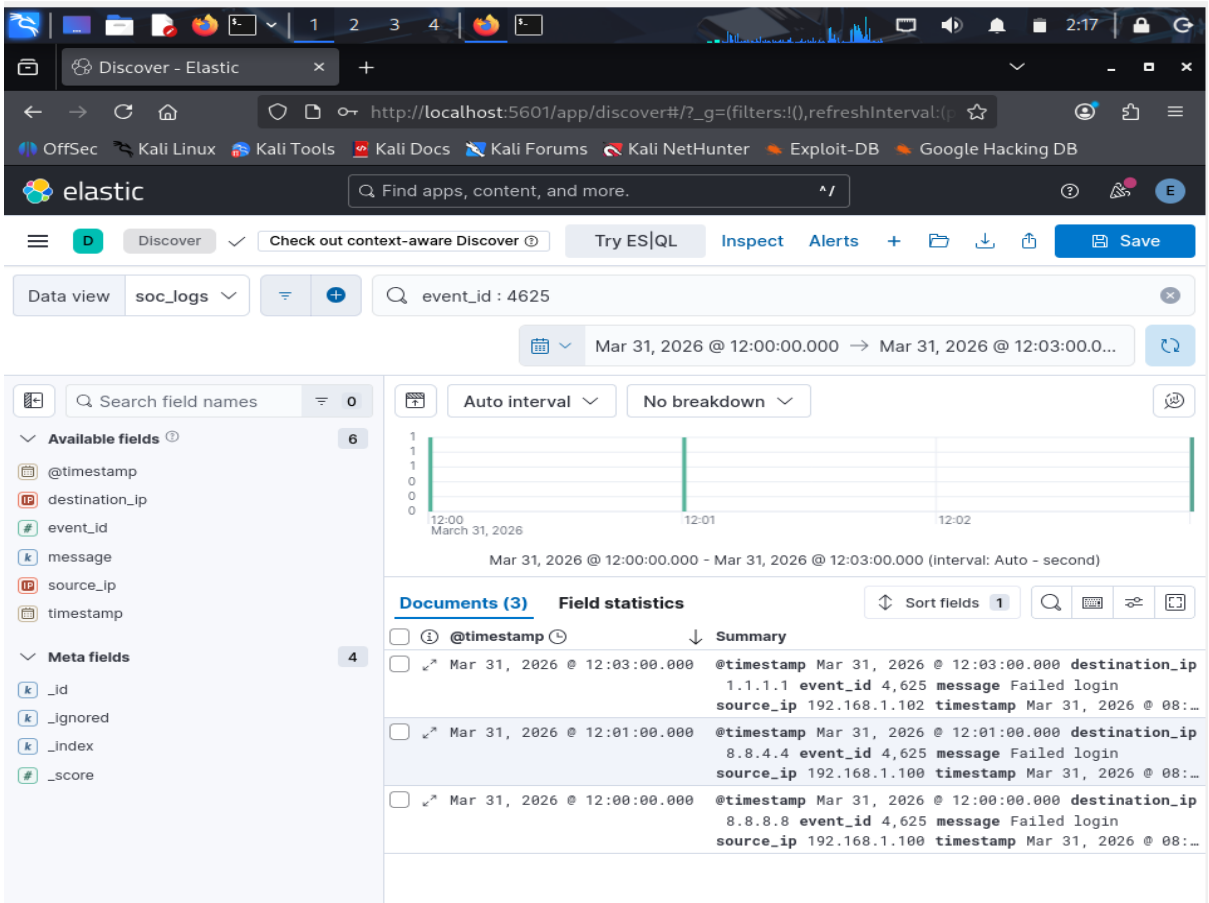

Fig: 1- event_id: 4625

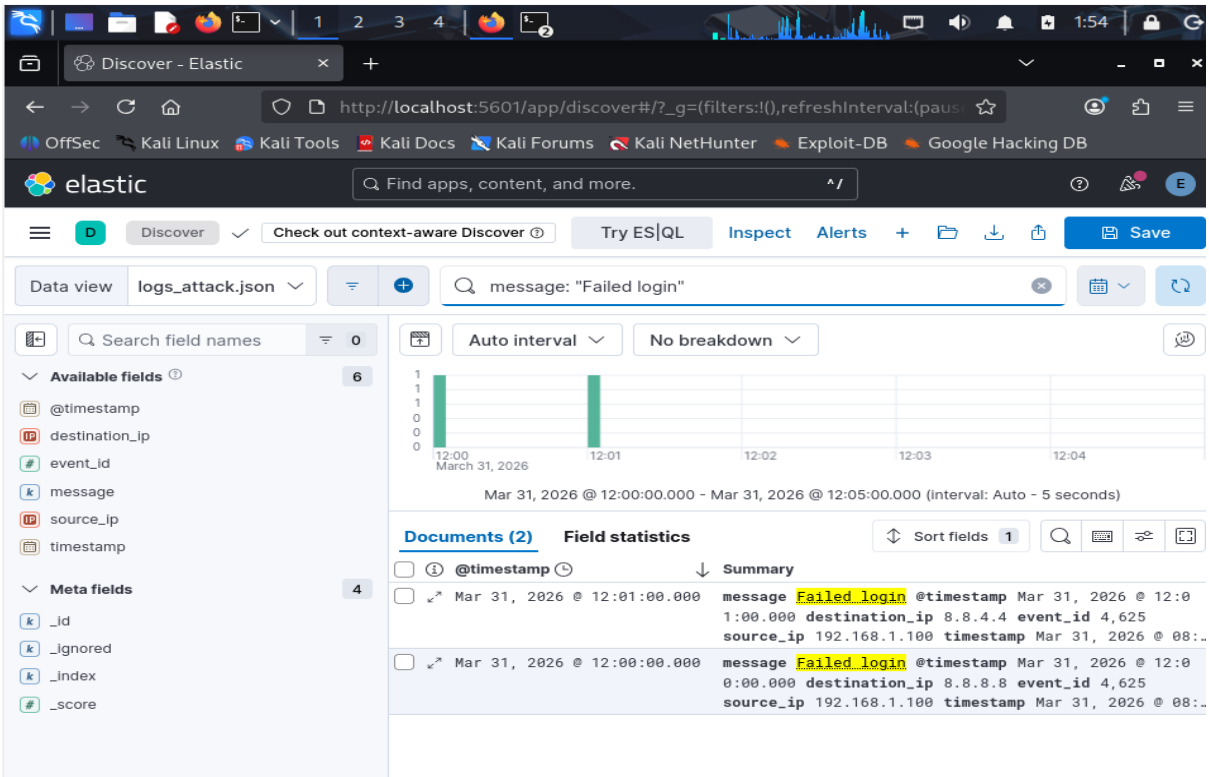

Fig: 2- message: "Failed Login"

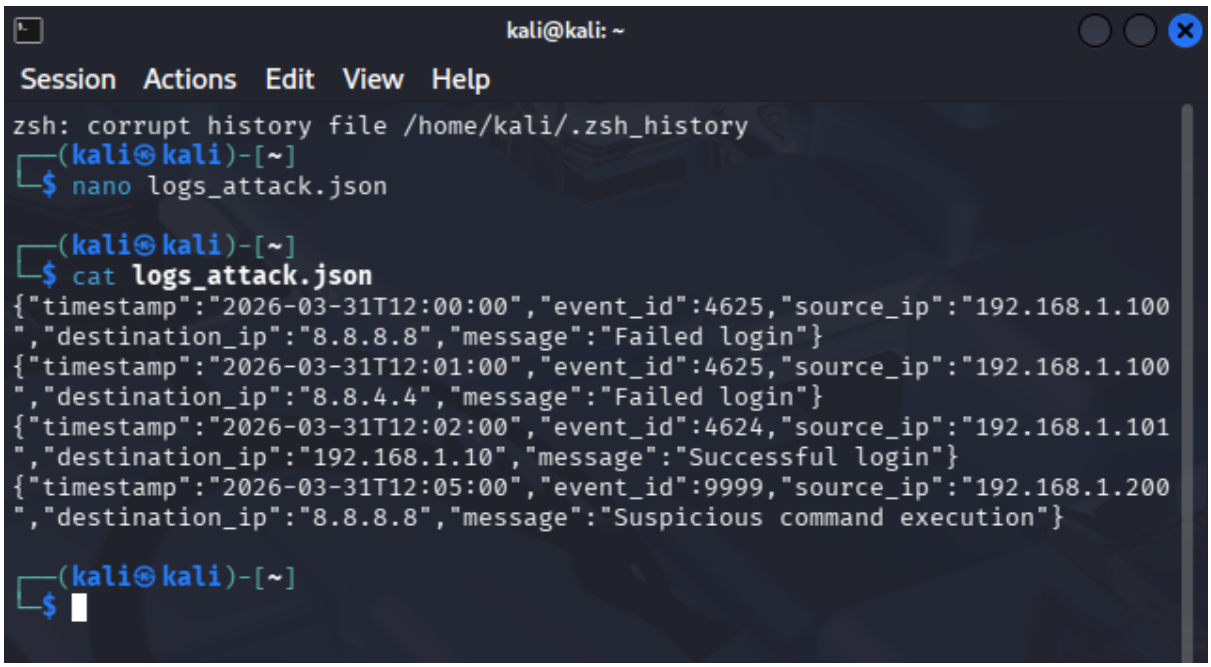

Fig: 3- Kibana_queries

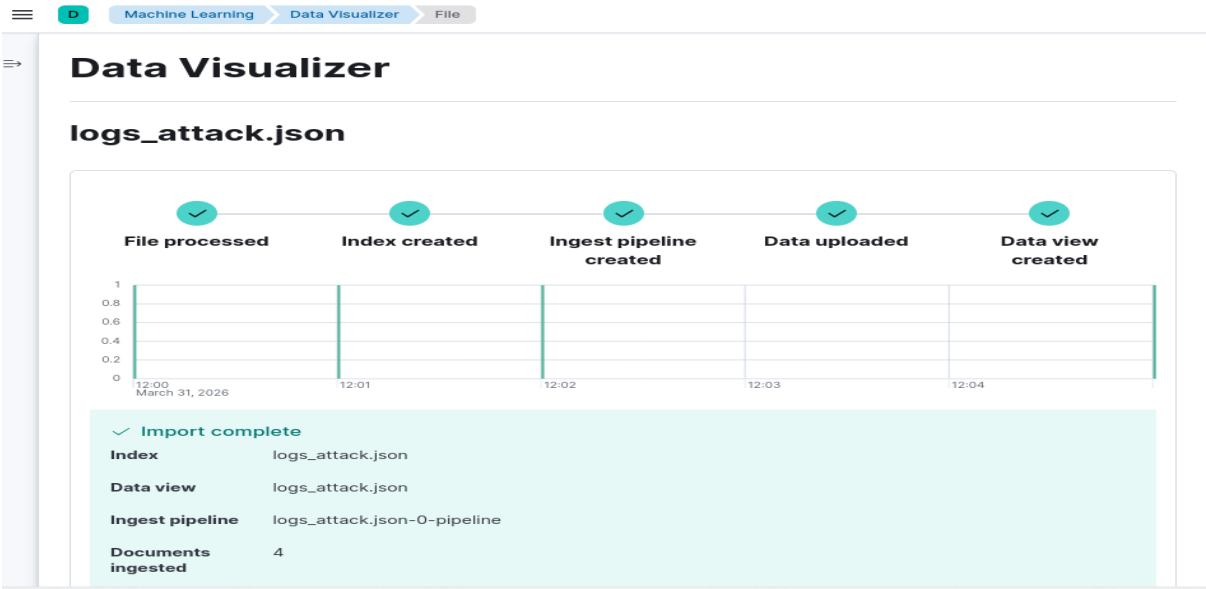

Fig: 4- logs_attack.json

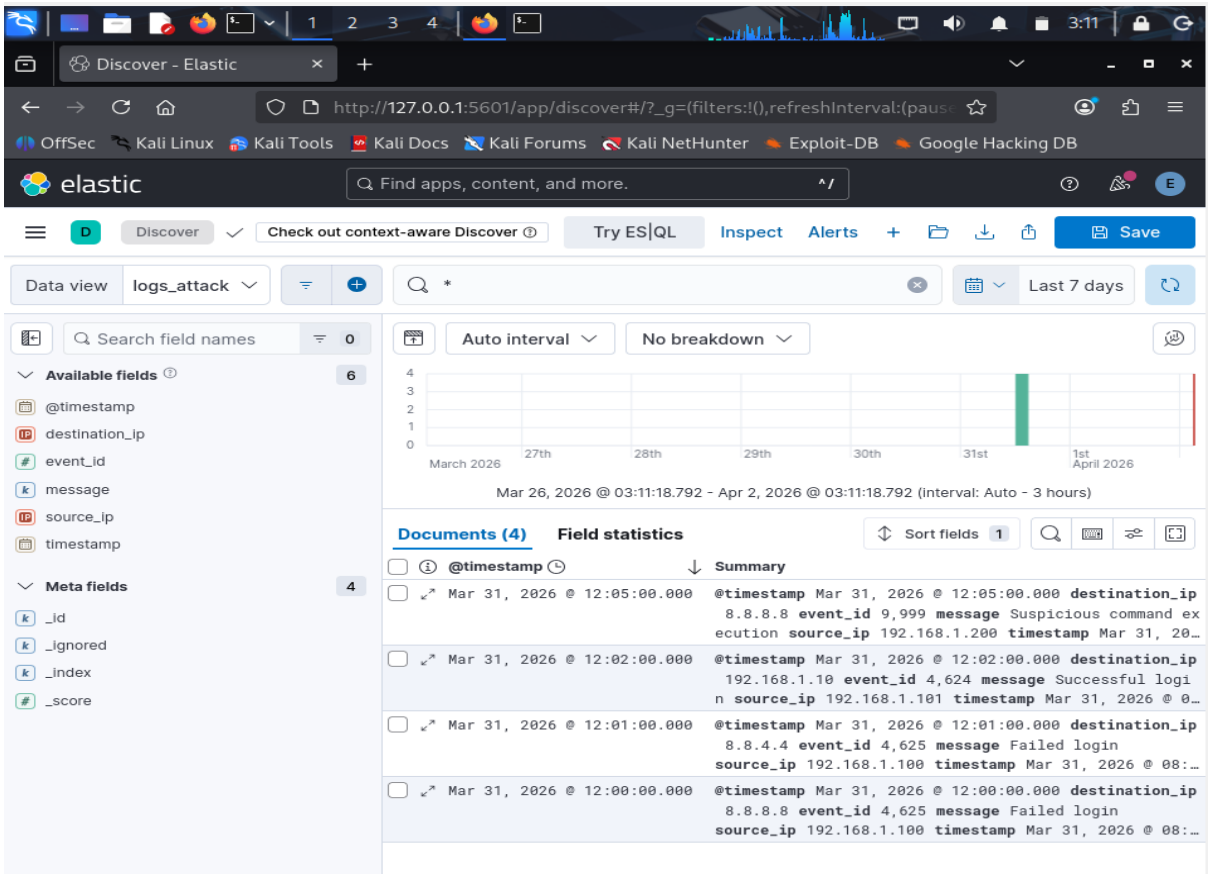

Fig: 5- Suspicious

- **Summary:-**

  Log analysis identified repeated failed logins and unusual traffic patterns. Correlation of events helped detect possible brute-force attempts. GeoIP enrichment provided additional context about IP locations, improving visibility. These findings indicate suspicious behavior requiring further investigation and response.

7. Threat Intelligence Integration:-

- ➢ In this step, I used threat intelligence tools like VirusTotal to check whether an IP address is malicious.
- ➢ I analyzed IP addresses from logs and compared them with threat databases. This helps in identifying whether the activity is known malicious or safe.

- **Alert Enrichment Table:-**

Alert ID	IP	Reputation	Notes
001	8.8.8.8	Clean	Public DNS

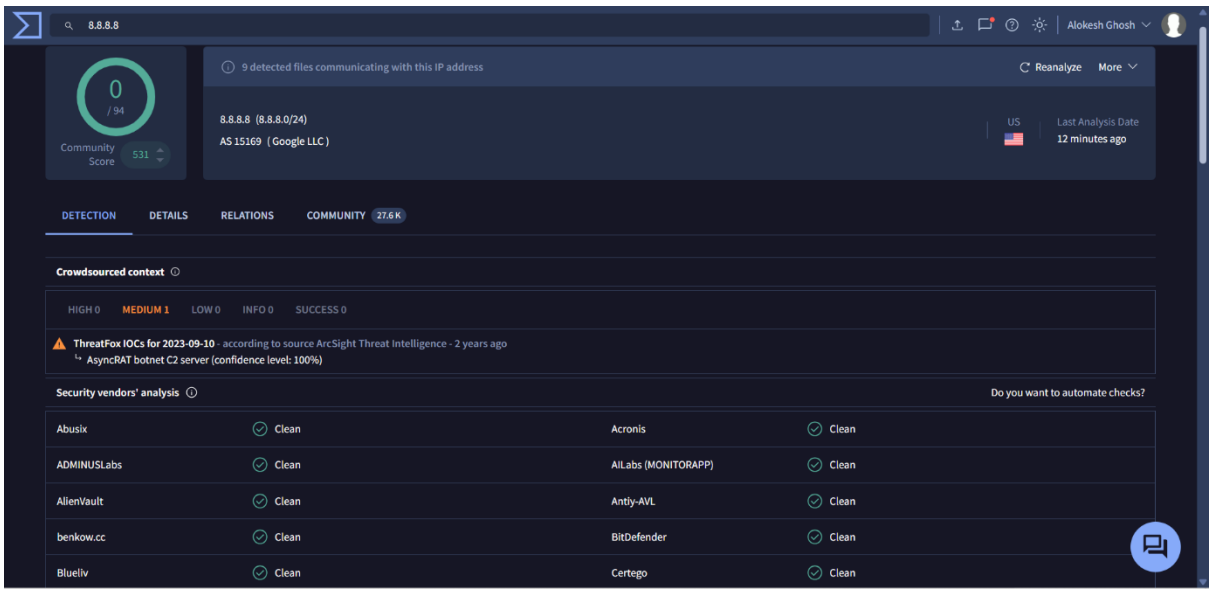

Fig: 6- Ip: 8.8.8.8

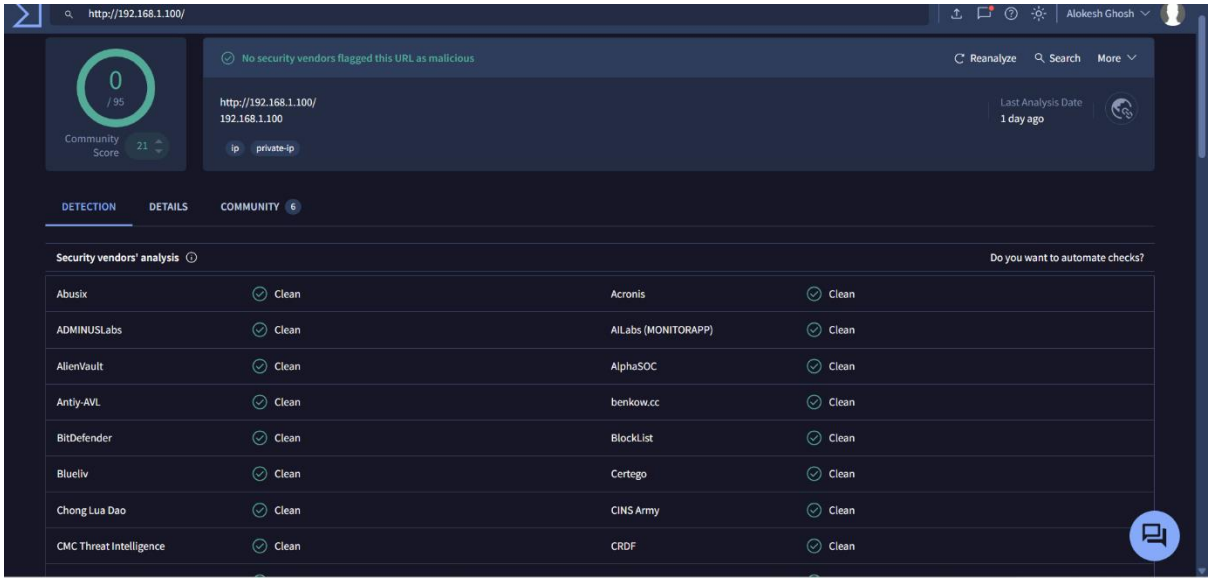

Fig: 7- Ip: 192.168.1.100

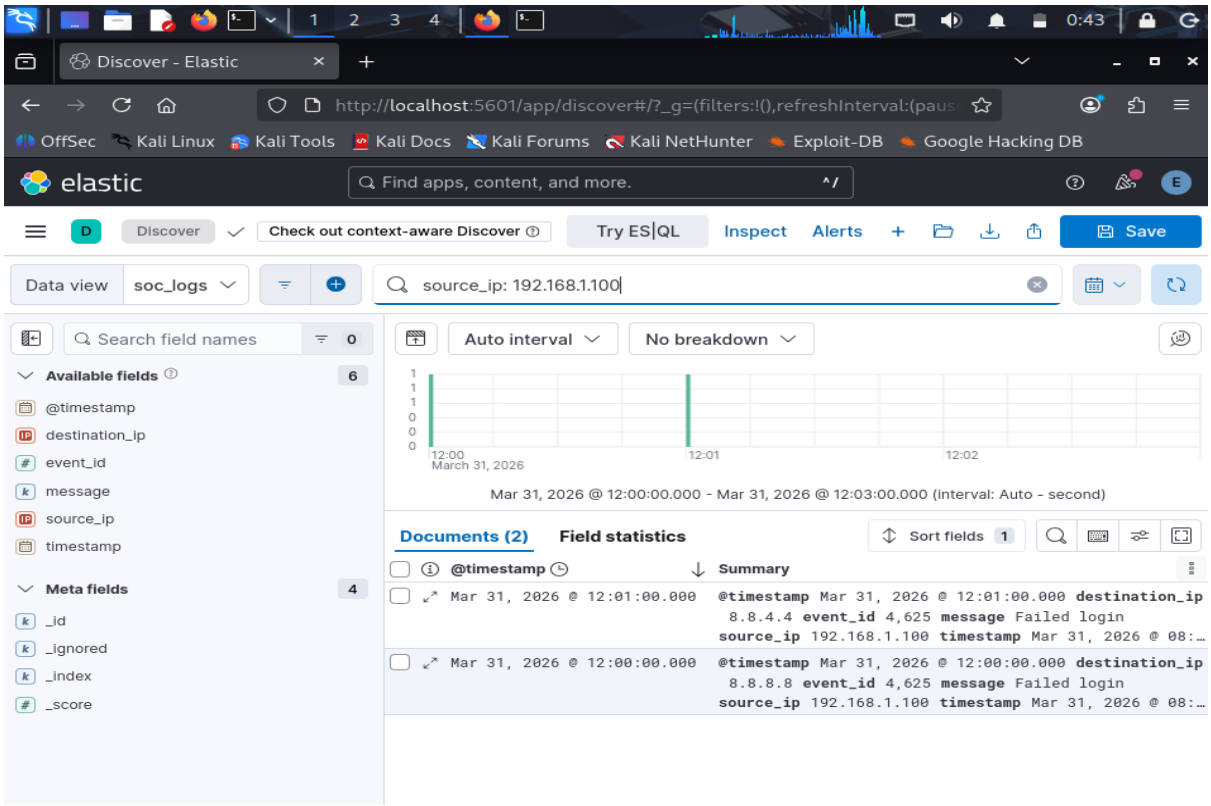

Fig: 8- source_ip

- **Summary:-**

  Threat intelligence validation was performed using VirusTotal. The analyzed IP showed no malicious activity but provided context from community data. This process helps in verifying alerts and reducing false positives, improving decision-making in SOC operations.

8. Incident Escalation Practice:-

In this task, I simulated an incident escalation process.

A high-priority alert was identified and escalated to Tier 2. A proper summary was written to explain the issue clearly.

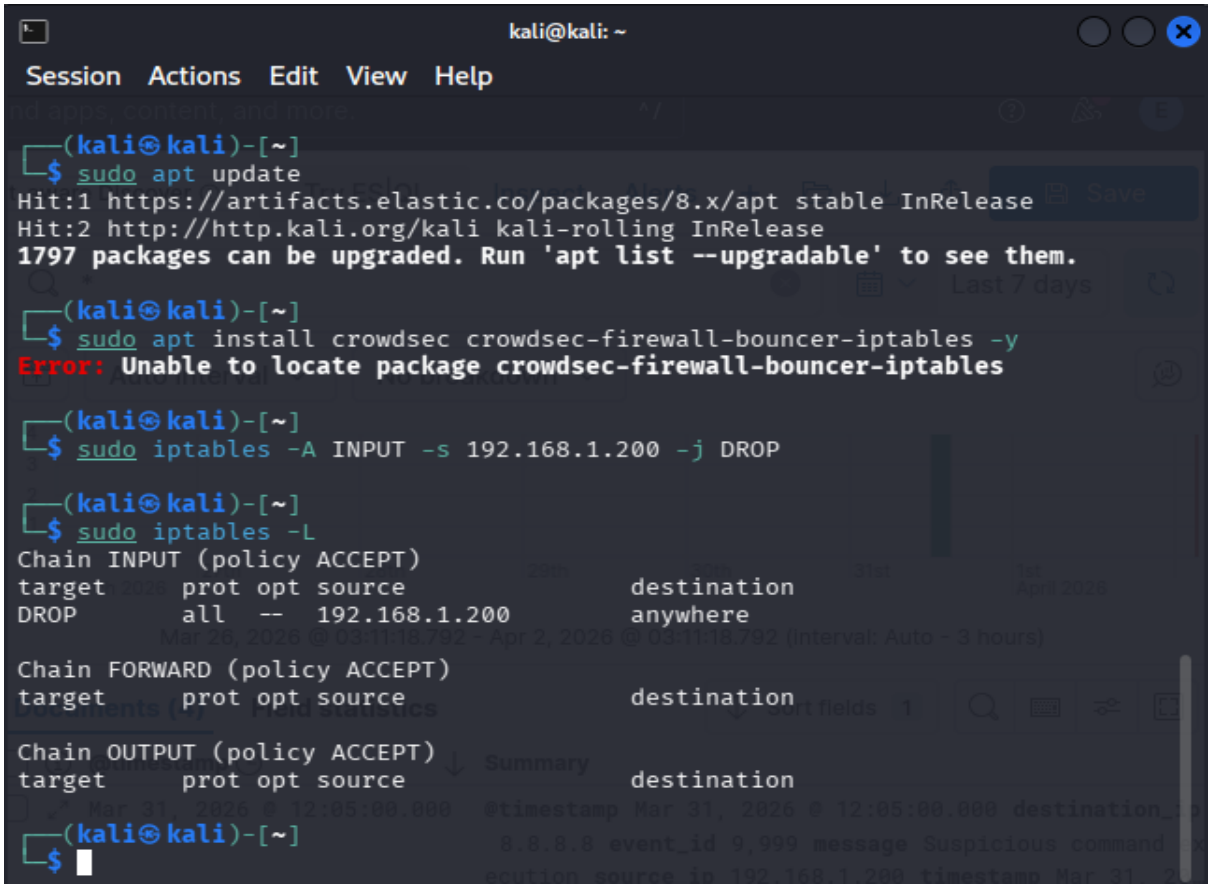

Fig: 9- commands.txt

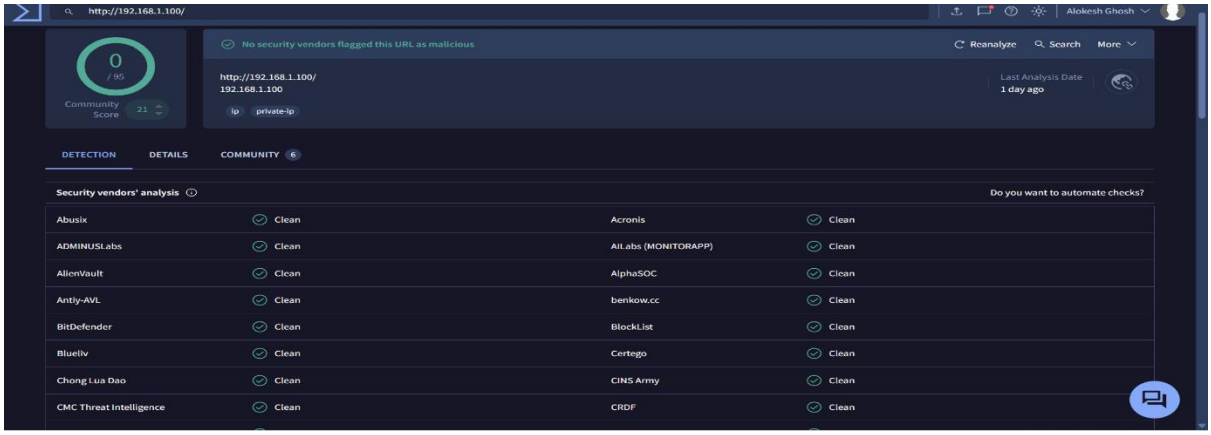

Fig: 10- Ip 192.168.1.100

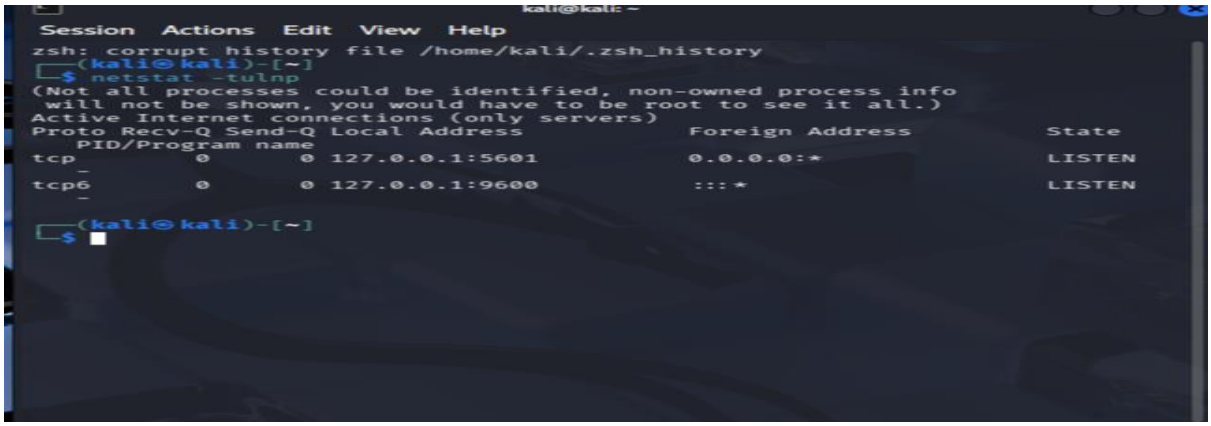

Fig: 11- netsat – tulnp

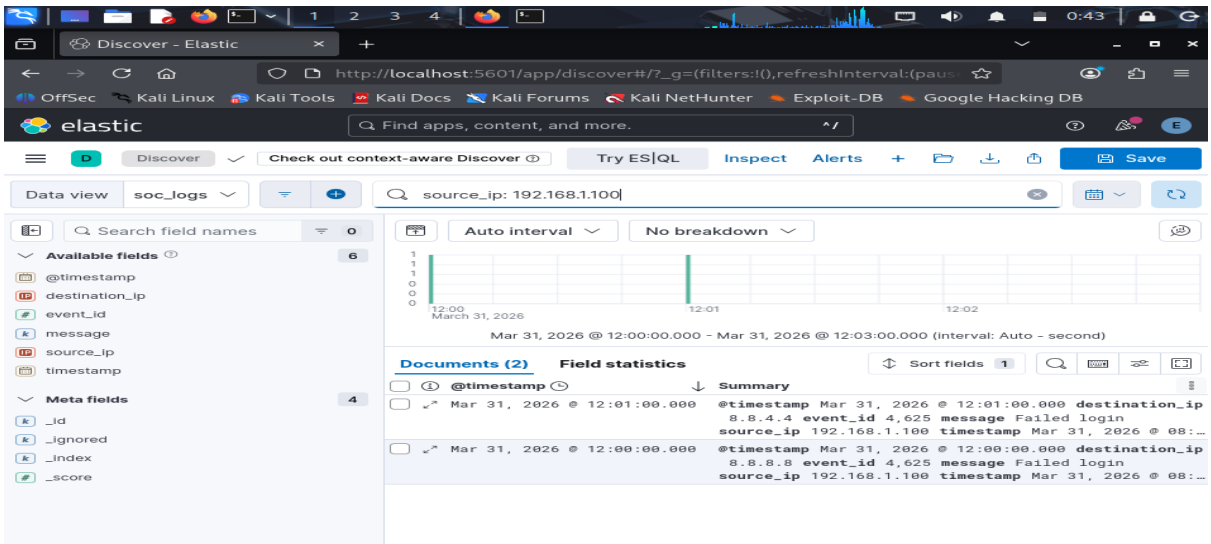

Fig: 12- souce_ip

- **Escalation Summary:-**

  A high-priority alert was triggered due to unauthorized access detected on Server-Y. The activity originated from IP 192.168.1.200 and indicates possible credential misuse. The system was isolated to prevent further access. Logs were collected for analysis. Due to the severity of the incident, it was escalated to Tier 2 for deeper investigation. Further analysis is required to identify lateral movement and prevent further compromise.

9. Alert Triage with Threat Intelligence:-

In this step, I selected a suspicious alert and analyzed it.

The alert showed unusual activity such as suspicious command execution. The IP was verified using VirusTotal to check its reputation.

- **Triage Table:-**

Alert ID	Description	Source IP	Priority	Status
004	Suspicious command execution	192.168.1.200	High	Open

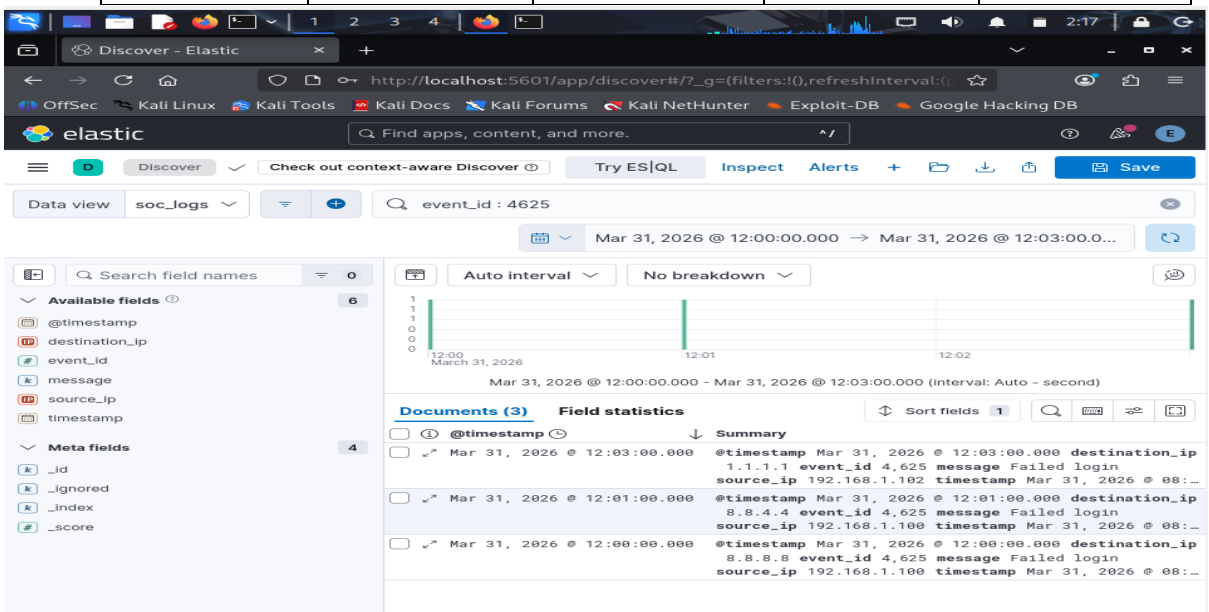

Fig: 13- event_id 4625

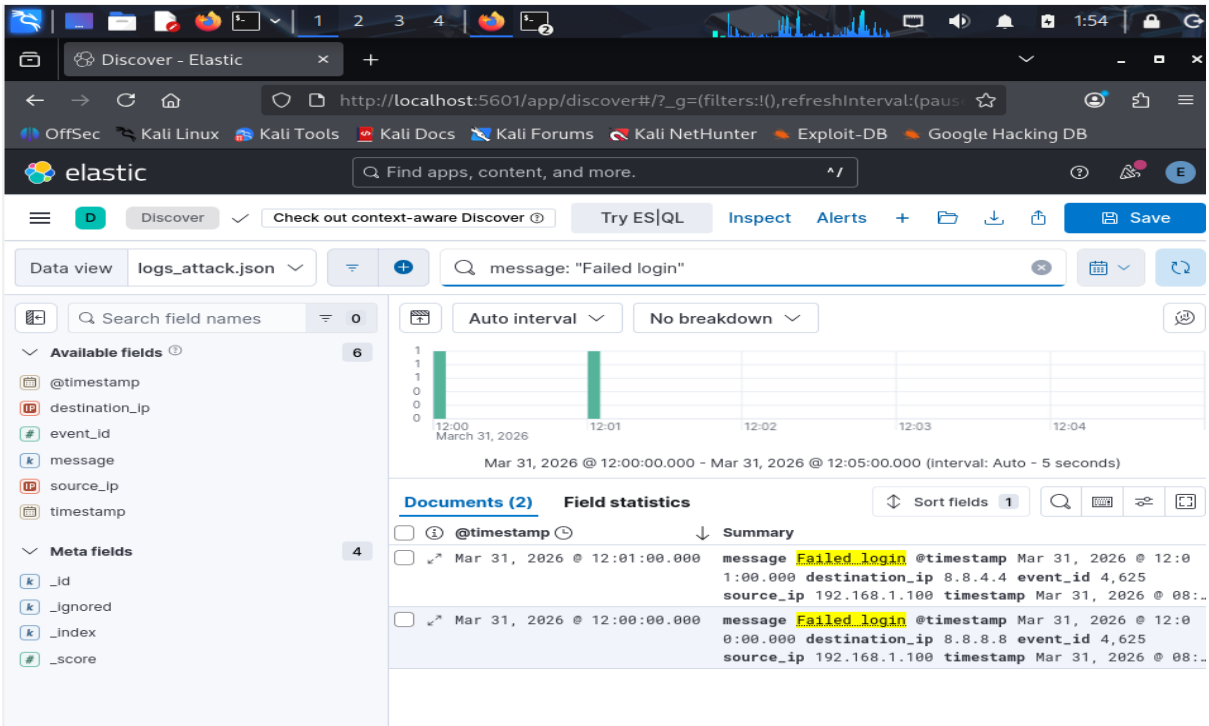

Fig: 14- Failed_login

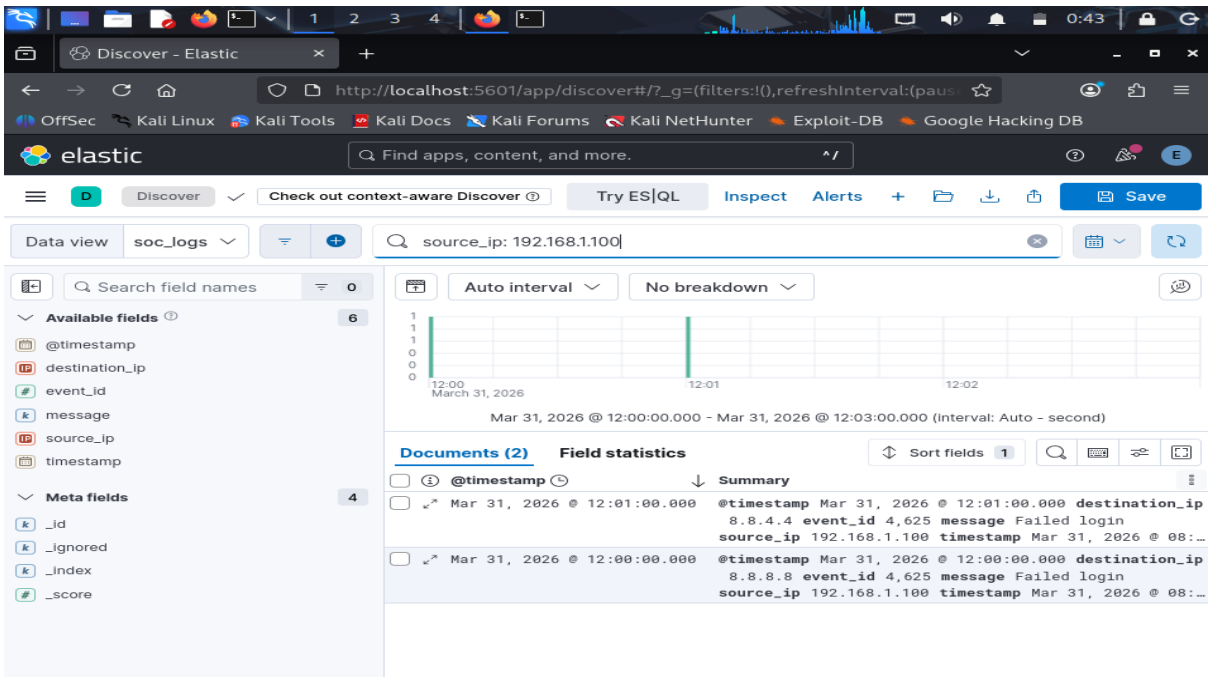

Fig: 15- source_ip

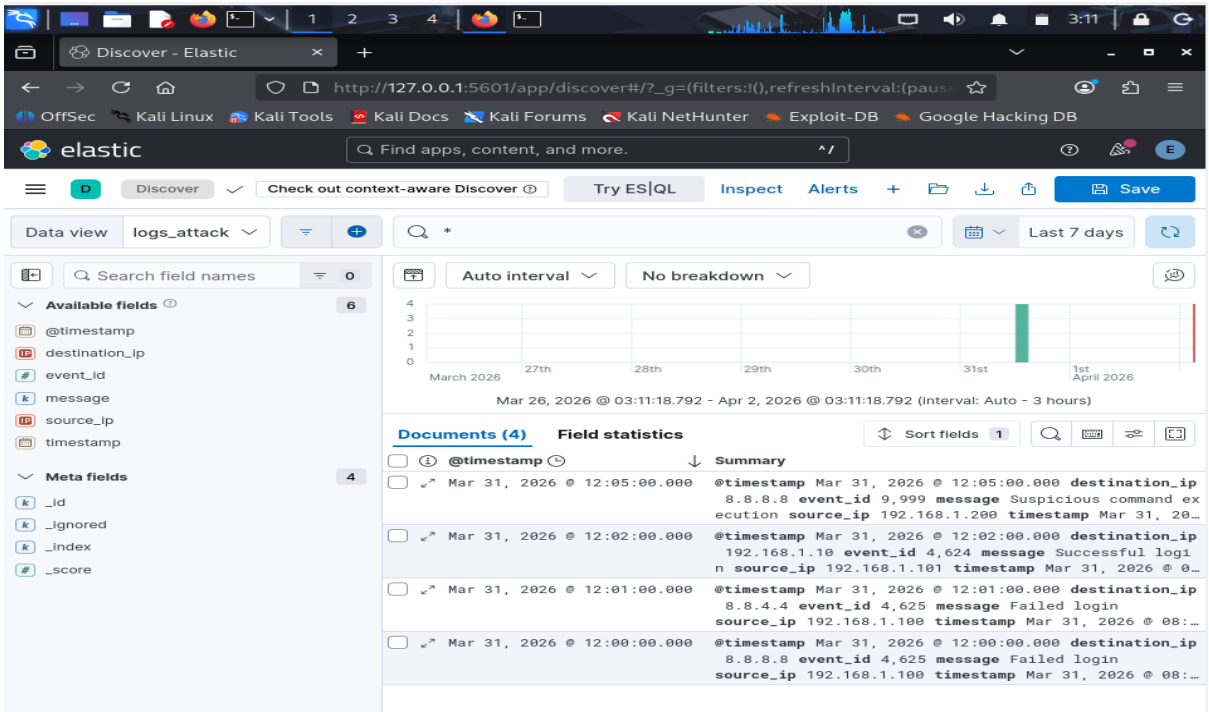

Fig: 16- Suspicious

- **Summary:-**

  The alert indicated suspicious command execution from an internal IP. Threat intelligence validation provided additional insights into the activity. Based on analysis, the alert was marked as high priority and required immediate investigation to prevent further system compromise.

10.Evidence Preservation and Analysis:-

In this task, I collected system data using terminal commands.

Network connections were captured and saved as a CSV file. Hashing was performed to maintain data integrity.

- **Evidence Table:-**

Item	Description	Collected By	Date	Hash Value
Network data	Netstat output	SOC Analyst	2026-03-31	SHA256...

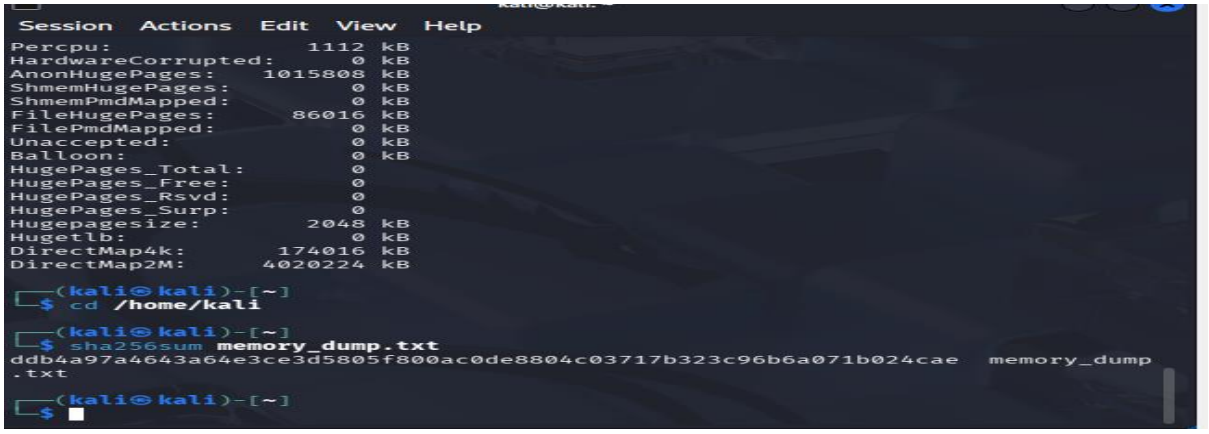

Fig: 17- hash.txt

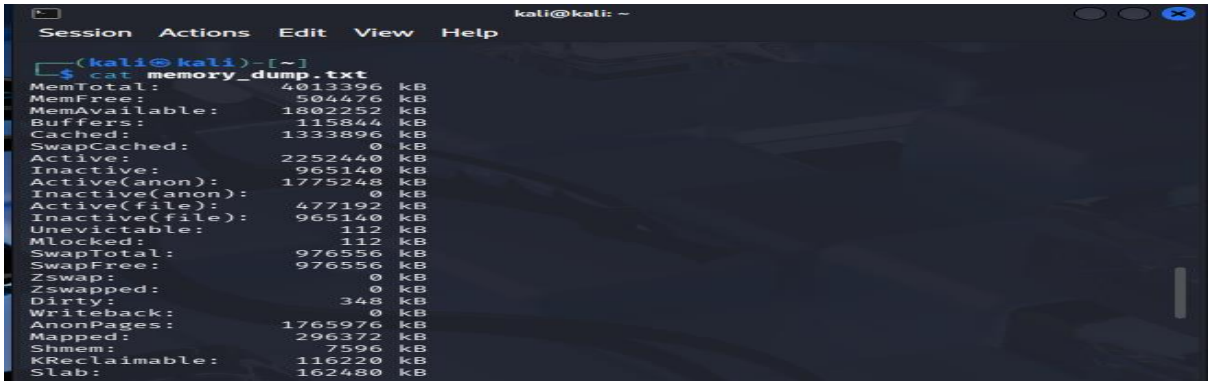

Fig: 18- memory_dump

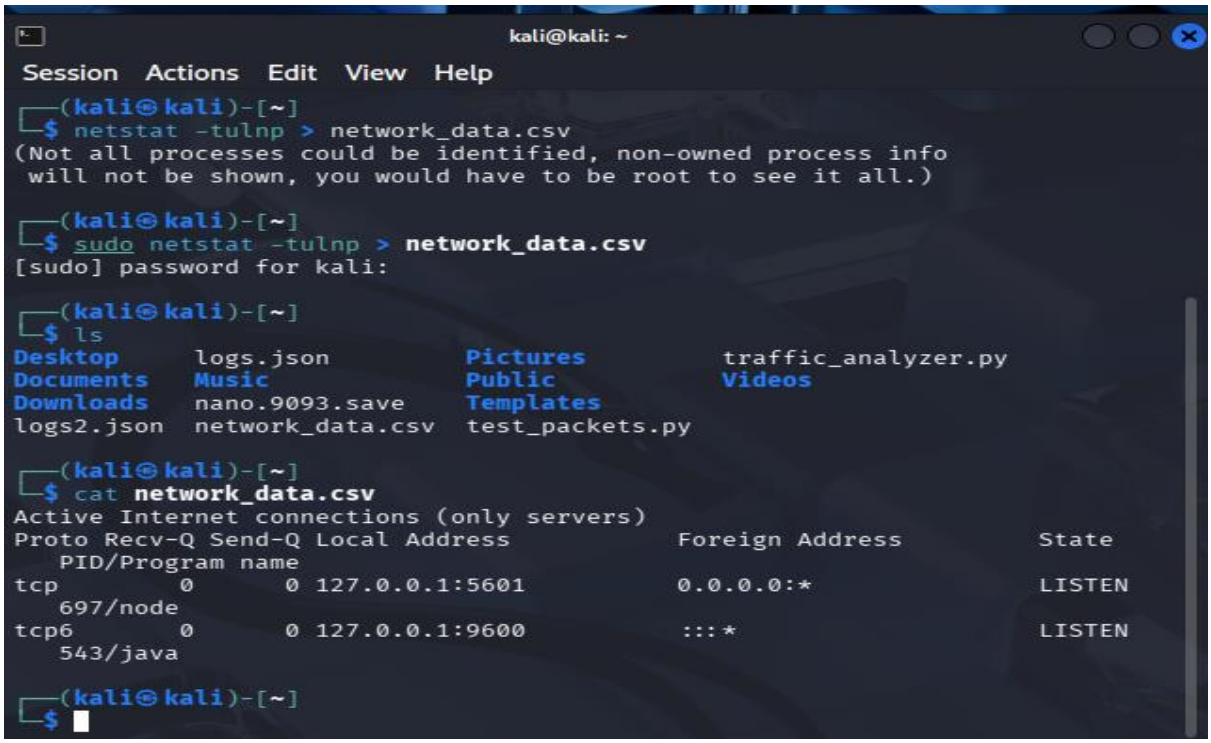

Fig: 19- network_data.csv

11. Capstone Project:-

- In this final step, I simulated a full SOC workflow including detection, analysis, and response.
- Suspicious activity was detected, analyzed, and mapped to MITRE ATT&CK technique T1210.

➢ **Detection Table:-**

Timestamp	Source Ip	Alert Description	MITRE Technique
2026-03-31 12:05:00	192.168.1.200	Suspicious execution	T1210

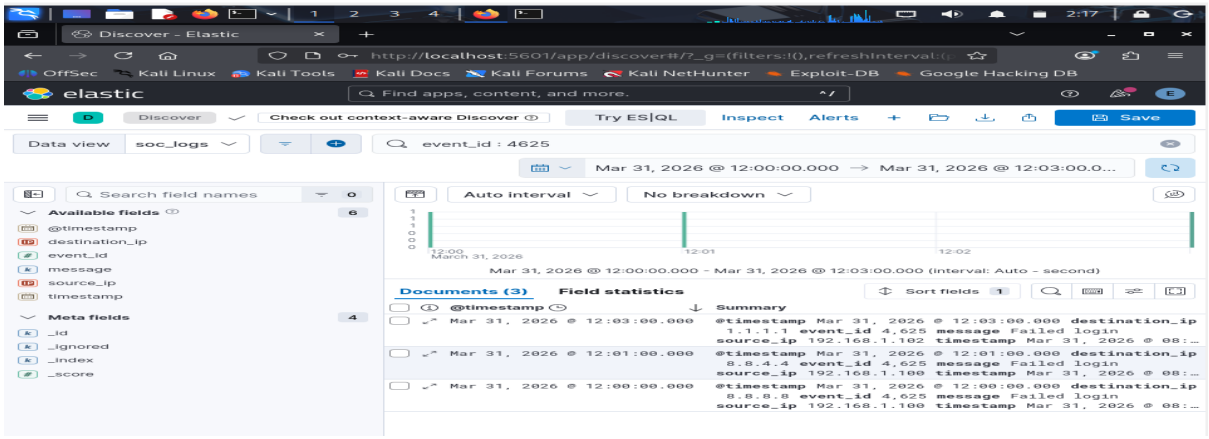

Fig: 20- event_id 4625

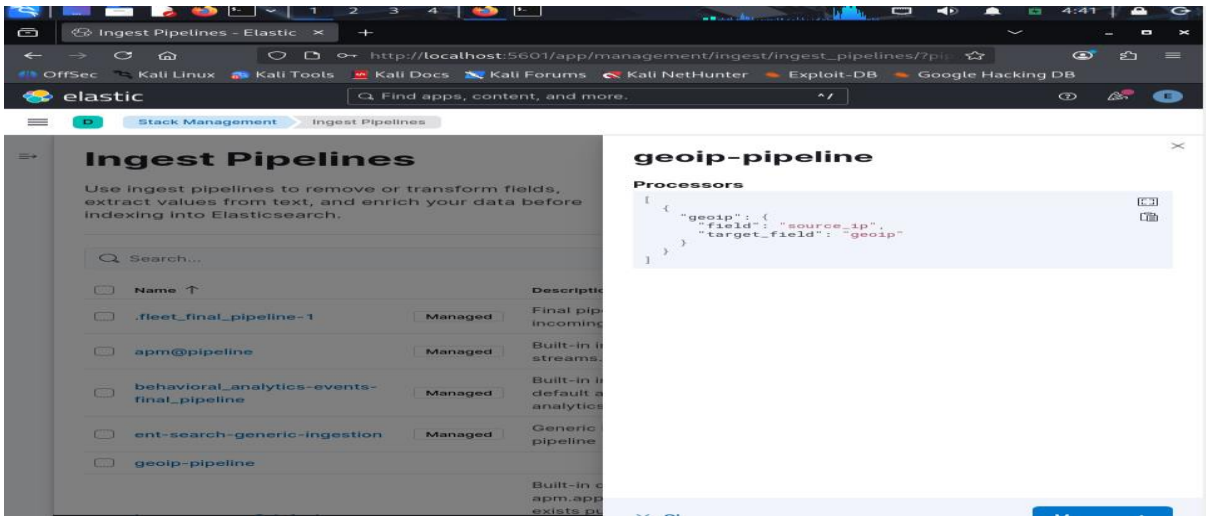

Fig: 21- geoip-pipeline

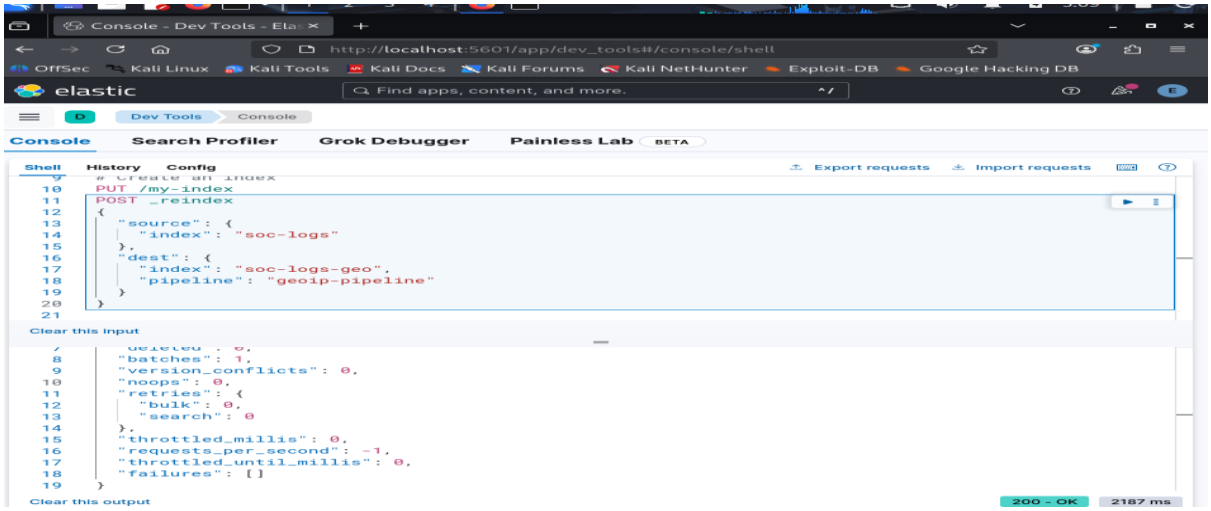

Fig: 22- geoip-pipeline-code

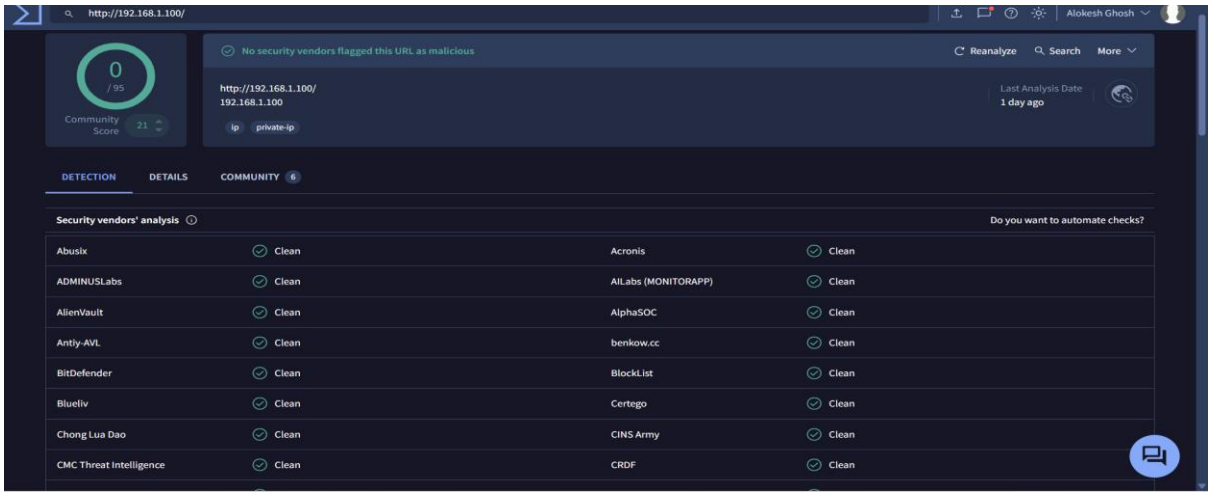

Fig: 23- ip: 192.168.1.100

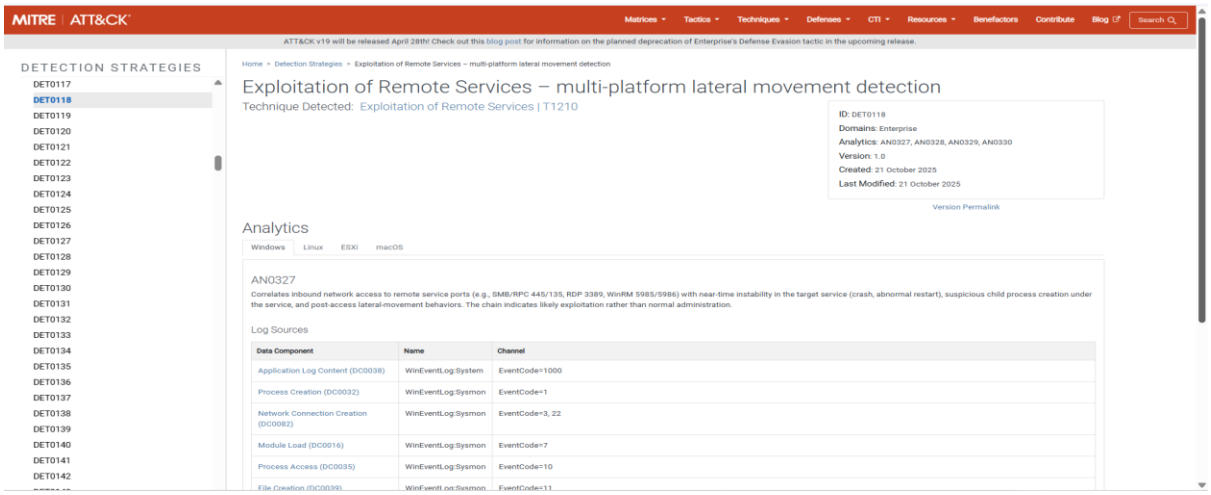

Fig: 24- MITRE_MAPPING

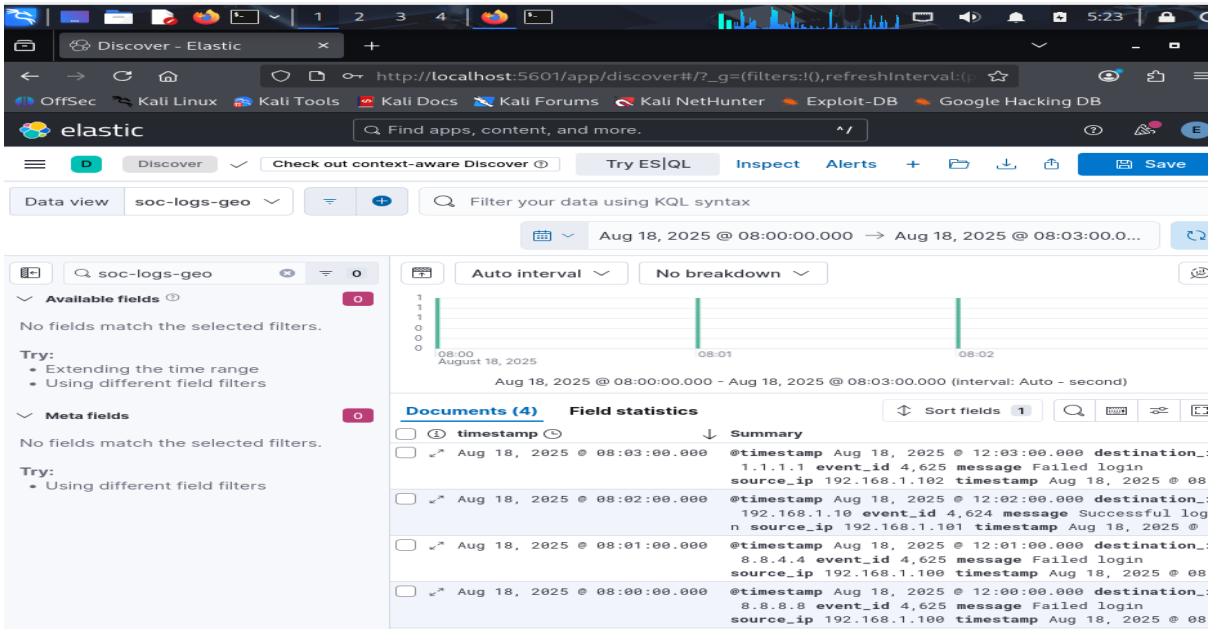

Fig: 25- soc-logs-geo

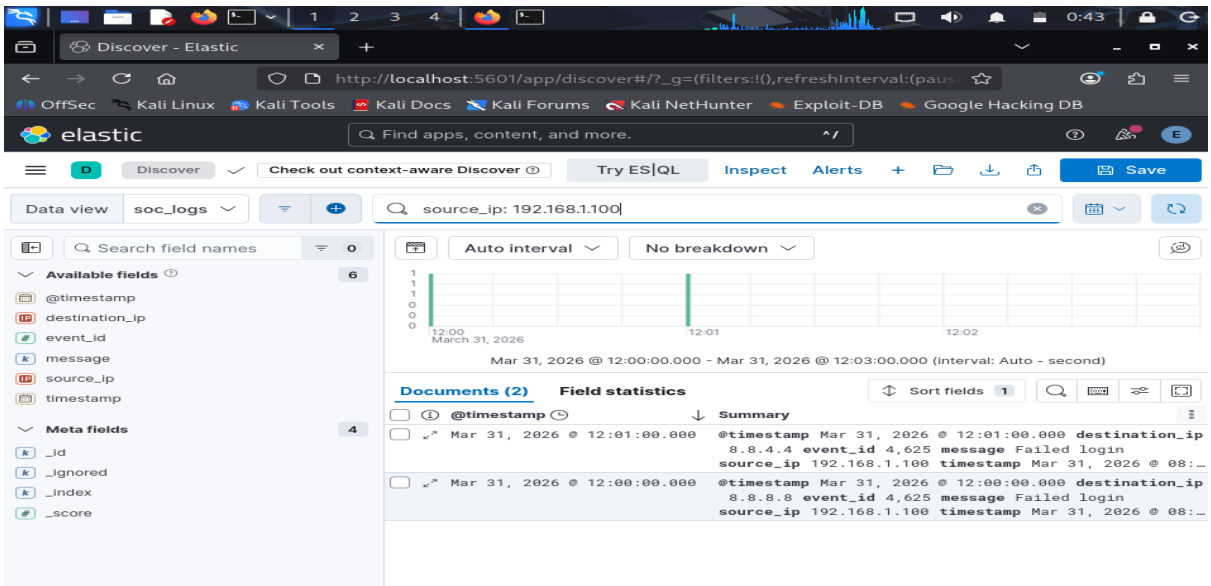

Fig: 26- source_ip

12. Final Report:-

- The project simulated a real-world SOC workflow where suspicious activity was detected and analyzed.
- Log analysis identified unauthorized access attempts and command execution. Threat intelligence tools were used to validate the IP reputation.
- The incident was triaged and marked as high priority. The affected system was isolated and the attacker IP was blocked.

- MITRE ATT&CK mapping identified the attack as T1210. The incident was escalated for further investigation.
- Proper documentation and reporting were maintained throughout the process.
- This project demonstrates practical SOC skills including detection, response, and reporting.

13. Conclusion:-

- This practical work helped in understanding real SOC operations. It improved skills in log analysis, threat intelligence, incident response, and reporting.
- These skills are essential for detecting and handling cyber threats effectively.